

CS & IT ENGINEERING



'C' Programming

Pointers & Arrays

Lecture No.- 03



By- Satya sir

Recap of Previous Lecture



- Pointer datatype
- Pointer size
- Void Pointer
- NULL Pointer
- Constant Pointer
- Arrays
 - 1-D array

Topics to be Covered



- Initialization of 1-D array
- Array Programs
- String handling in 'C'





Topic : 1-D Array



1-D Array Initialization

- Along with declaration

Ex: `int x[5] = {10, 20, 30};`
`printf("%d", x[3]); // 0`
`printf("%d", x[4]); // 0`

	0	1	2	3	4
x	10	20	30	0	0

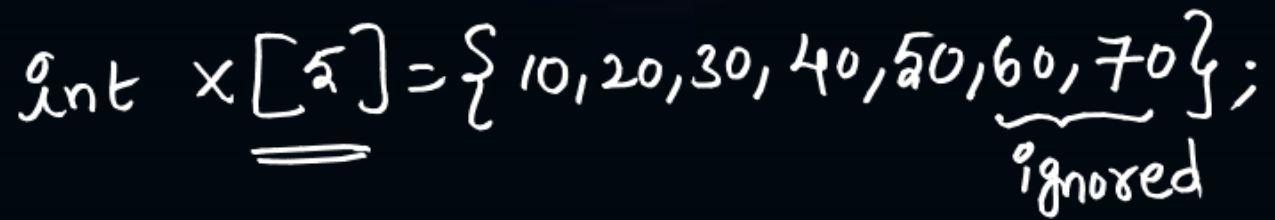
`int x[5];`

`printf("%d %d %d", x[0], x[2], x[4]);`

// Garbage values

`int x[5] = {0};`

	0	1	2	3	4
x	0	0	0	0	0



$\text{int } x[] = \{10, 20, 30, 40, 50, 60, 70\}$
 size $\rightarrow 7$

```
int x[];  
// Error, array size must be declared
```

```
int x[5] = {10, 20, 30};
```

`printf("%d", x[6]);`
`x[2] = x[3] + x[7];`

Access leads to error.



Topic : String Handling in 'C'



// Find Minimum Element from array

```
#include <stdio.h>
```

```
int main( )
```

```
{ int x[20], n, min, i;
```

```
printf("Enter No. of Elements");
```

```
scanf("%d", &n);
```

```
for(i=0; i<n; i++)
```

```
scanf("%d", &x[i]);
```

```
min = x[0];
```

```
for(i=1; i<n; i++)
```

```
{ if(x[i] < min)
```

```
{ min = x[i];
```

```
printf("The minimum element is %d", min);
```

```
return 0;
```

```
}
```

For Maximum Element

→ if (x[i] > max)

max = x[i];



Topic : String Handling in 'C'



Strings :

- Group of characters (or) character array (or) character pointer also.

Syntax : `char stringname[size];`

`char x[20];`

Initialization : `char x[] = {'G', 'A', 'T', 'E', 'E', 'x', 'A', 'M'};`

(OR)
`char x[] = "GATEEXAM";`

0	1	2	3	4	5	6	7
G	A	T	E	E	x	A	M

X

0	1	2	3	4	5	6	7	8
G	A	T	E	E	x	A	M	\0

X

NULL
Character

Ex: `char str[5];`

`printf("%c", str[0]);`
`printf("%c", str[2]);`

} Garbage values

`char str[5] = {'x'};`

`printf("%c %c", str[2], str[4]);`

o/p: No output

0	1	2	3	4
x	\0	\0	\0	\0

str



Topic : String Handling in 'C'



```
char str[5] = {'C'};
```

	0	1	2	3	4
str	C	\0	\0	\0	\0

```
printf(" %.d %.d %.d", str[0], str[2], str[4]);
```

o/p: 67 0 0
 = ↓ ↓
 'C' ASCII \0 ASCII
 value value

```
int x[5];
```

```
for(i=0; i<5; i++)  
  scanf(" %.d", &x[i]);
```

```
char x[5];
```

```
scanf(" %.s", x);
```

```
int x[5], y[5];
```

```
for(i=0; i<5; i++)  
  scanf(" %.d %.d", &x[i], &y[i]);
```

```
char x[5], y[5];
```

```
scanf(" %.s %.s", x, y);  
// Accepts only 1 i/p.
```



Topic : String Handling in 'C'



strlen()

Syntax: strlen(string)

strlen(): returns the index of first NULL character (`\0`) if exists, otherwise size of string.

char x[5] = {'A', 'B', 'C', 'D', 'E'};

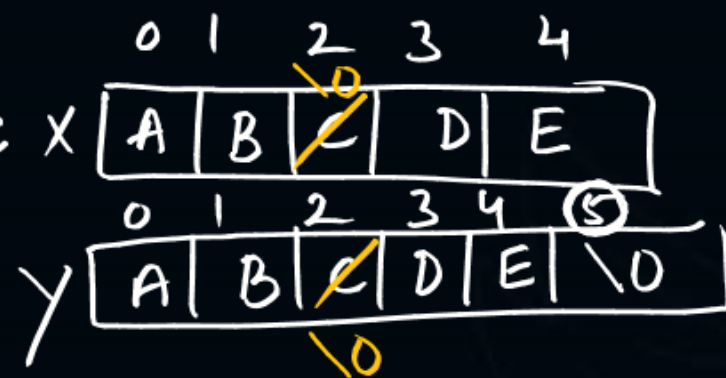
char y[5] = "ABCDE";

printf("%d", strlen(x)); #5

printf("%d", strlen(y)); #5

printf("%d", sizeof(x)); #5

printf("%d", sizeof(y)); #6



`\0` Indicates End of String

sizeof()

⇒ Returns the amount of memory allocated in Bytes.

x[2] = '\0'; y[2] = '\0';

printf("%d", strlen(x)); #2

printf("%d", strlen(y)); #2

printf("%d %d", sizeof(x), sizeof(y));

// 2 6



Topic : String Handling in 'C'



No.	Function	Description
1)	<code>strlen(string_name)</code>	returns the length of string name.
2)	<code>strcpy(destination, source)</code>	copies the contents of source string to destination string.
3)	<code>strcat(first_string, second_string)</code>	concatenates or joins first string with second string. The result of the string is stored in first string.
4)	<code>strcmp(first_string, second_string)</code>	compares the first string with second string. If both strings are same, it returns 0.
5)	<code>strrev(string)</code>	returns reverse string.
6)	<code>strlwr(string)</code>	returns string characters in lowercase.
7)	<code>strupr(string)</code>	returns string characters in uppercase.
8)	<code>strstr(str1, str2)</code>	It returns a pointer to the first occurrence of the given substring str2 within the given string str1



Topic : String Handling in 'C'



Please read and go through

No.	Function	Description
9)	strncmp()	It compares two strings only to n characters.
10)	strncat()	It concatenates n characters of one string to another string.
11)	strncpy()	It copies the first n characters of one string into another.
12)	strchr()	It finds out the first occurrence of a given character in a string.
13)	strrchr()	It finds out the last occurrence of a given character in a string.
14)	strnstr()	It finds out the first occurrence of a given string in a string where the search is limited to n characters.
15)	strcasecmp()	It compares two strings without sensitivity to the case.
16)	strncasecmp()	It compares n characters of one string to another without sensitivity to the case.



Topic : String Handling in 'C'



- strlen()
- strcat()
- strcmp()
- strlwr()
- strupr()
- strchr()
- strcpy()



2 mins Summary



- 1-D array Initialization
- Strings
- String vs Number arrays

To be Contd 



THANK - YOU